

Understanding Machine Learning and Deep Learning Through Real-World Examples

Introduction

Machine learning (ML) falls under the broader field of artificial intelligence, and deep learning (DL) is one area within machine learning. Both use data to help computers recognize patterns and make decisions, but they differ in how they learn from and process data. In machine learning, people usually identify the most important features before training the model. Deep learning uses multilayered neural networks to discover useful features directly from raw data. The most suitable method depends on the type of data, the difficulty of the task, and the amount of computing power available.

Machine Learning Example: House Price Prediction

One machine learning example from the lesson is using linear regression to predict house prices. This can be seen on real estate websites that show an estimated value for a home. The estimate is based on details such as the location, square footage, number of bedrooms, year the house was built, and the prices of similar homes nearby. The model studies past sales and looks for patterns between these details and the final selling price. It then uses those patterns to estimate the value of another house.

Machine learning is suitable for this task because house price data is usually organized into clear categories, such as location, size, number of rooms, and the age of the property. These factors can be selected before the model is trained, making linear regression a suitable choice. It works well with structured data, requires less computing power, and is easier to understand than a deep learning model. Deep learning would not be the best choice because it usually needs much more data and processing power. It would also make the prediction process more complicated without providing a clear advantage for this type of data.

Deep Learning Example: Autonomous Driving

One deep learning example from the lesson is autonomous driving. A real world application of this approach can be seen in self-driving cars. These vehicles use cameras, sensors, and other devices to collect information about their surroundings. The deep learning model analyzes this information to recognize roads, traffic signs, pedestrians, other vehicles, and obstacles. It then uses what it has learned from large amounts of driving data to help the vehicle make safe driving decisions.

Deep learning is suitable for this task because autonomous driving requires the vehicle to process large amounts of complex data from cameras and sensors in real time. Deep learning models can automatically learn important features from this data, allowing them to

recognize roads, traffic signs, pedestrians, other vehicles, and obstacles with high accuracy. Machine learning would be less suitable because it usually depends on manually selected features, making it more difficult to handle the complexity and changing conditions involved in autonomous driving.

Conclusion

Machine learning and deep learning can both be useful but they are not equally suited for every problem. Machine learning works well when the data is organized into clear categories, as shown in house price prediction. Deep learning is more useful when the system must process complex information, such as images and sensor data in autonomous vehicles. The better choice depends on the kind of data being used and the difficulty of the task. Knowing the difference between these approaches makes it easier to select the right method for a particular problem.